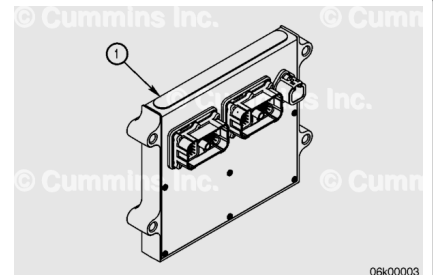


Trainings ▾

General Information

This procedure applies **only** to engines equipped with electronically controlled injectors.



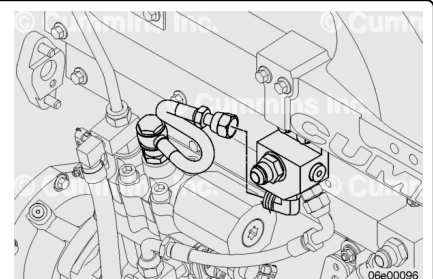
Fuel Pressure Relief Valve Return Flow Test

Obtain a graduated cylinder that is marked in cubic centimeters; example: graduated cylinder, Part Number 4919139, or equivalent.

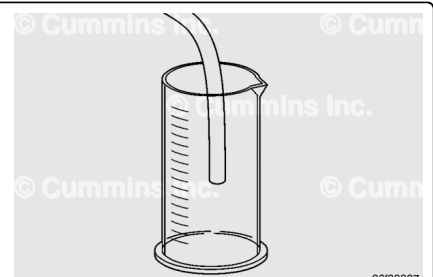


Remove the fuel pressure relief valve drain line from the fuel drain manifold.

Install an M18 plug in the fuel drain manifold.

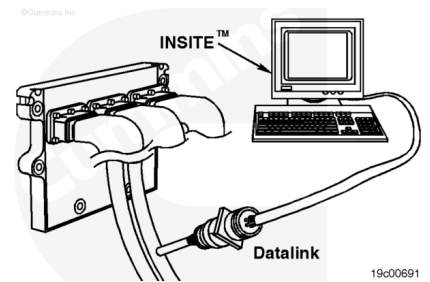


Route the fuel pressure relief valve drain line into a graduated cylinder.



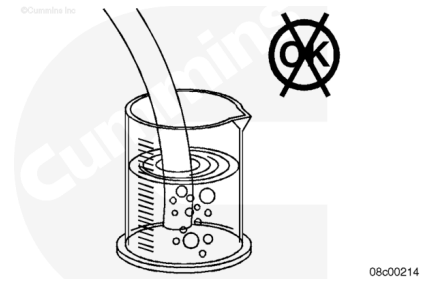
Connect Cummins® electronic service tool or equivalent, start engine, and operate the Fuel System Leakage Test.

See Cummins® electronic service tool or equivalent manual for instructions.



Leakage **must** be less than 10 drops per minute.

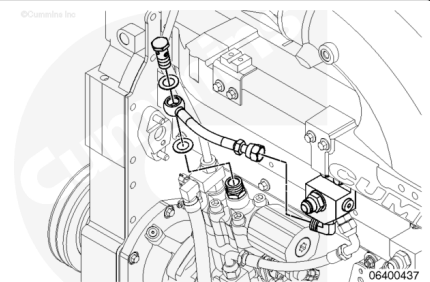
A steady leak from the fuel pressure relief valve can be an indication that the valve seat is worn.



Install the fuel pressure relief valve drain line into the fuel drain manifold.

Torque Value:

Banjo Capscrews 48 n•m [35 ft-lb]



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (1) battery cable last.



⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe

personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

A very small amount of dirt and debris can be very harmful to the fuel pump. Extra care is required to keep the fuel connections clean during removal and installation. Connections must be covered immediately to keep them clean when components are removed from the fuel pump.

- Disconnect the battery cables. See equipment manufacturer service information.
- Clean the fuel pump and surrounding area, and dry with compressed air.
- Disconnect the fuel drain line from the fuel pump. Use a clean plastic plug to protect the fuel drain line port.
 - For K38 and K50 engines, use the following procedure in the K38 and K50 Service Manual, Bulletin 4021528. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/28/28-006-013.html)
 - For QSK38 and QSK50 engines, use the following procedure in the QSK38 and QSK50 Service Manual, Bulletin 4021530. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-013-tr.html)
 - For QSK50 CM2350 K108 and QSK50 CM2670 K176C engines, use the following procedure in the

QSK50 CM2350 K108 and QSK50 CM2670 K176C
Service Manual, Bulletin 4332823. Refer to
Procedure 006-013 in Section 6.

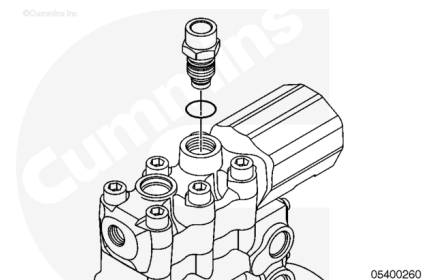
(/qs3/pubsys2/xml/en/procedures/255/255-006-013-tr.html)

- For QSK78 CM2350 K126 and QSK78 CM2670 K184C engines, use the following procedures in the QSK78 CM2350 K126 and QSK78 CM2670 K184C Service Manual, Bulletin 5504266. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/461/461-006-013.html)
- For QSK78 CM2350 K137 and QSK78 CM2670 K185C engines, use the following procedures in the QSK78 CM2350 K137 and QSK78 CM2670 K185C Service Manual, Bulletin 5504636. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/591/591-006-013.html)

Remove

The fuel pump is equipped with metric fasteners. Use metric size tools for the removal and installation of the valve.

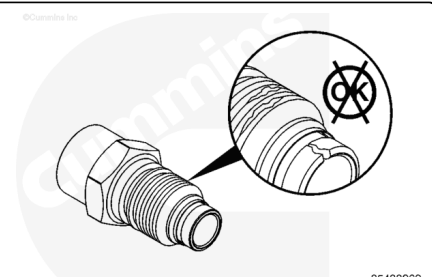
Remove the mechanical dump valve from the fuel pump.
Discard the o-ring.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean all parts with electrical contact cleaner, Part Number 3824510, or equivalent.

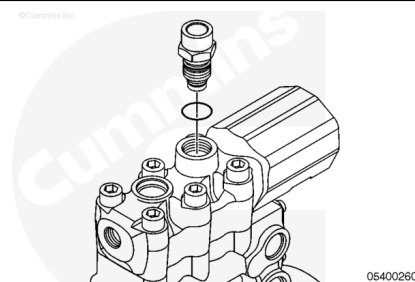
Inspect the mechanical dump valve for stripped threads, damage, restricted passage, or cracks. Replace the valve, if damaged.

Install

Install the new o-ring on the mechanical dump valve.

Install the mechanical dump valve in the fuel pump.

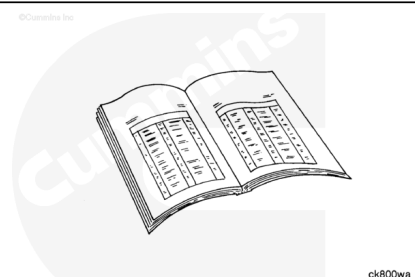
Torque Value: 135 n•m [100 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (1) battery cable last.



- Install the fuel return line.
 - For K38 and K50 engines, use the following procedure in the K38 and K50 Service Manual, Bulletin 4021528. Refer to Procedure 006-013 in Section 6.
(</qs3/pubsys2/xml/en/procedures/28/28-006-013.html>)
 - For QSK38 and QSK50 engines, use the following procedure in the QSK38 and QSK50 Service Manual, Bulletin 4021530. Refer to Procedure 006-013 in Section 6.

(/qs3/pubsys2/xml/en/procedures/56/56-006-013-tr.html)

- For QSK50 CM2350 K108 and QSK50 CM2670 K176C engines, use the following procedure in the QSK50 CM2350 K108 and QSK50 CM2670 K176C Service Manual, Bulletin 4332823. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/255/255-006-013-tr.html)
- For QSK78 CM2350 K126 and QSK78 CM2670 K184C engines, use the following procedures in the QSK78 CM2350 K126 and QSK78 CM2670 K184C Service Manual, Bulletin 5504266. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/461/461-006-013.html)
- For QSK78 CM2350 K137 and QSK78 CM2670 K185C engines, use the following procedures in the QSK78 CM2350 K137 and QSK78 CM2670 K185C Service Manual, Bulletin 5504636. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/591/591-006-013.html)
- Connect the battery cables. See equipment manufacturer service information.

Start and operate the engine and check for leaks.